

Supplementary material for: "Fair to Federate? Two-Sided Prediction of Performance and Fairness Outcomes in Healthcare Federated Learning"

Anonymous Author(s)

A Dataset Details

ChestX-ray14 is a large dataset of labeled radiographs from tens of thousands of patients. We focus on seven pathological conditions commonly used as benchmarks in chest X-ray analysis, namely Atelectasis, Cardiomegaly, Consolidation, Edema, Effusion, Pneumonia, and Pneumothorax. The dataset exhibits a moderate sex imbalance, with male patients representing 56% of the data.

Fitzpatrick17k is a dataset of skin lesion images annotated with skin tone via the Fitzpatrick scale. We binarize skin tone into light-skin ($\{1, 2, 3, 4\}$) and dark-skin ($\{5, 6\}$) groups, the latter constituting a small minority of the data ($\approx 14\%$). The classification target is derived from the coarsest level of the skin condition taxonomy, which partitions images into benign neoplastic, malignant neoplastic, and non-neoplastic conditions; we merge the two neoplastic categories into a single positive class (tumoral conditions), with non-neoplastic conditions as the negative class.

Fitzpatrick17k further provides two independent skin tone annotations per image, which do not always agree due to the inherent subjectivity of skin tone labeling. We use this to distinguish between *reliable* samples, where both annotators agree on the binarized group assignment, and *unreliable* ones, where they disagree or either annotation is missing. Reliable samples are prioritized when constructing the validation and test sets to ensure that fairness evaluation rests on consistent and trustworthy labels; unreliable samples may appear in client training sets once the reliable supply for a given budget is exhausted.

B Model Architecture and Training Details

In the chest X-ray setting, all models use DenseNet-121 [2] pretrained on ImageNet, with the classification head replaced by a 1024×7 fully connected layer and sigmoid activation, trained with binary cross-entropy applied independently to each pathology output. Images are resized and center-cropped to 256×256 pixels, normalized using ImageNet statistics, and augmented during training with random horizontal flips and rotations up to 10° .

In the dermatology setting, all models use EfficientNet-B2 [4] pretrained on ImageNet, with the classifier replaced by a single sigmoid output unit. To address class imbalance, binary cross-entropy with logits is used alongside class weighting and a weighted random sampler. Images are resized to 320 pixels, randomly cropped to 288×288 , and augmented with flips, rotations, and color jitter. In both settings, optimization is performed with Adam.

C Justification for the Sampling Budget

The choice of 800 federated models per multi-client setting is motivated by two complementary considerations.

First, it ensures that the two-client, three-client, and five-client datasets contribute a comparable number of observations to the

pooled regression dataset, preventing any single federation scale from dominating the fit.

Second, 800 experiments distributed across the 27 candidate-client strata yield approximately 30 observations per stratum. Treating the number of participants as a fixed offset leaves $d = 3$ effective predictors—total training size, weighted demographic majority proportion, and weighted flip fraction—and a standard rule of thumb recommends a minimum of 10 observations per predictor for stable coefficient estimation [1, 3], giving $10 \times 3 = 30$ per stratum. This reasoning applies directly to the parametric models in our suite; for the non-parametric ones, the low dimensionality and expected smoothness of the descriptor-outcome relationships substantially limit their effective complexity, and cross-validation further regularizes it, making 30 observations per stratum empirically sufficient across all model classes.

Finally, since outcomes sharing similar aggregate descriptor values across federation scales are expected to be highly correlated, the pooled dataset carries more information per stratum than three independent sets of 30 observations would, making the per-stratum budget of 30 a conservative rather than a tight choice.

D Additional results

D.1 Impact on the Candidate Client

Predicting federation gains. Figure 1 shows the actual versus predicted ΔBalAcc and ΔminTPR using the best-performing regression models, relatively to the analysis on the gains experienced by the candidate client upon joining the federation. Each point corresponds to a single federation configuration, with its predicted value obtained from the held-out fold in cross-validation, ensuring that no configuration is predicted by a model that has seen it during training. Predicted values lie close to the perfect-prediction diagonal across both datasets, indicating high predictive accuracy for these targets as well, and particularly so for the ΔminTPR .

Analyzing the most predictive features. Table 1 reports the mean drop in R^2 across five cross-validation folds when each feature group is removed from the dataset. The pattern of importances is consistent with that observed for the ΔAUC and the Results section: size-related features are strong predictors for the ΔBalAcc on NIH, while demographic composition and label bias features are the primary drivers in the remaining cases, with their relative importance being especially pronounced for Fitzpatrick17k. The only notable difference with respect to the results relative to ΔAUC and $\Delta\text{TPR}_{\text{gap}}$ is that label bias features cause a more pronounced drop in R^2 for the ΔBalAcc on NIH than size-related features.

SHAP beeswarm plots. Figure 2 presents SHAP beeswarm plots relative to the ΔBalAcc and ΔminTPR targets.

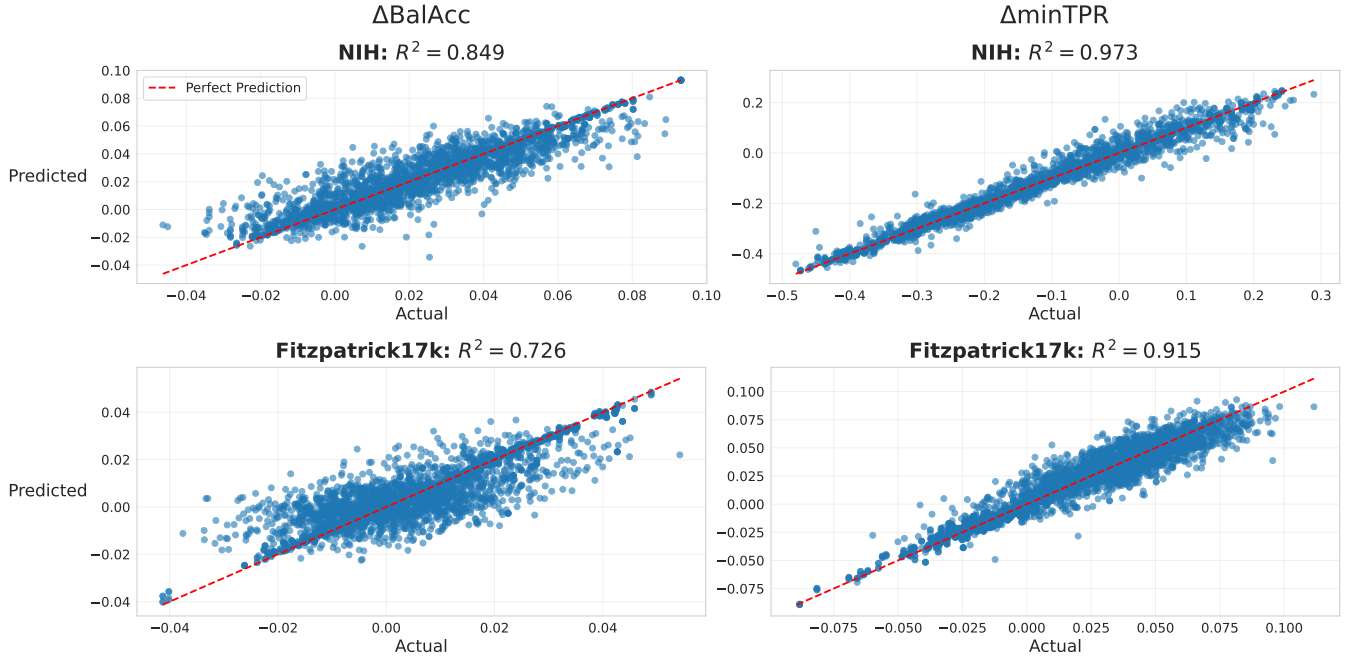


Figure 1: Actual vs. Predicted changes in BalAcc increment and minTPR increment experienced by the candidate client upon joining a federation, for NIH (top) and Fitzpatrick17k (bottom) datasets.

Table 1: Mean R^2 drop ($\pm$ standard deviation across five cross-validation folds) for each feature group and target, relative to NIH and Fitzpatrick17k. Results refer to the prediction of gains experienced by the candidate client upon joining a federation.

Feature group	NIH		Fitzpatrick17k	
	ΔBalAcc	ΔminTPR	ΔBalAcc	ΔminTPR
Size	0.148 ± 0.006	0.093 ± 0.09	0.008 ± 0.005	0.006 ± 0.001
Demographic composition	0.117 ± 0.007	0.126 ± 0.007	0.327 ± 0.025	0.146 ± 0.013
Label bias	0.239 ± 0.020	0.099 ± 0.012	0.277 ± 0.006	0.206 ± 0.015
Number of clients	0.008 ± 0.001	0.016 ± 0.004	0.000 ± 0.002	0.000 ± 0.001

Relatively to the ΔBalAcc on NIH, larger federation-to-candidate size ratios and smaller absolute candidate sizes are both associated with greater expected benefits. Higher label noise in the candidate is also associated with larger performance gains upon federation. For the minTPR increment, smaller and more demographically balanced candidate clients benefit more. Lower aggregate label bias in the federation is additionally associated with larger gains.

Relatively to the ΔBalAcc on Fitzpatrick17k, smaller candidate clients with a higher proportion of majority-group samples are associated with larger improvements. A more balanced demographic composition within the federation is also associated with greater gains.

For the ΔminTPR , higher flip fractions in the candidate are associated with larger improvements, as are more demographically balanced candidate clients. In contrast to the ΔBalAcc , the ΔminTPR increases with candidate client size.

Performance across federation sizes on Fitzpatrick17k. Figure 3 reports the R^2 values achieved by the best-performing model for each target, computed by filtering the held-out cross-validation folds to retain only configurations corresponding to two-, three- and five-client federations, respectively, for the Fitzpatrick17k setting.

For ΔAUC , predictive accuracy remains stable across federation sizes with no clear trend, oscillating between 0.83 and 0.87. For the remaining targets, results are broadly consistent with those observed on NIH: R^2 values slightly increase with federation size, rising from 0.66 to 0.74 for balanced accuracy increment, from 0.93 to 0.95 for soft $\Delta\text{TPR}_{\text{gap}}$, and from 0.87 to 0.93 for minimum soft TPR increment.

Taken together, these results confirm that federation gains remain reliably predictable across federation scales, and that representing the federation through aggregate descriptors rather than individual client characteristics does not degrade predictive performance.

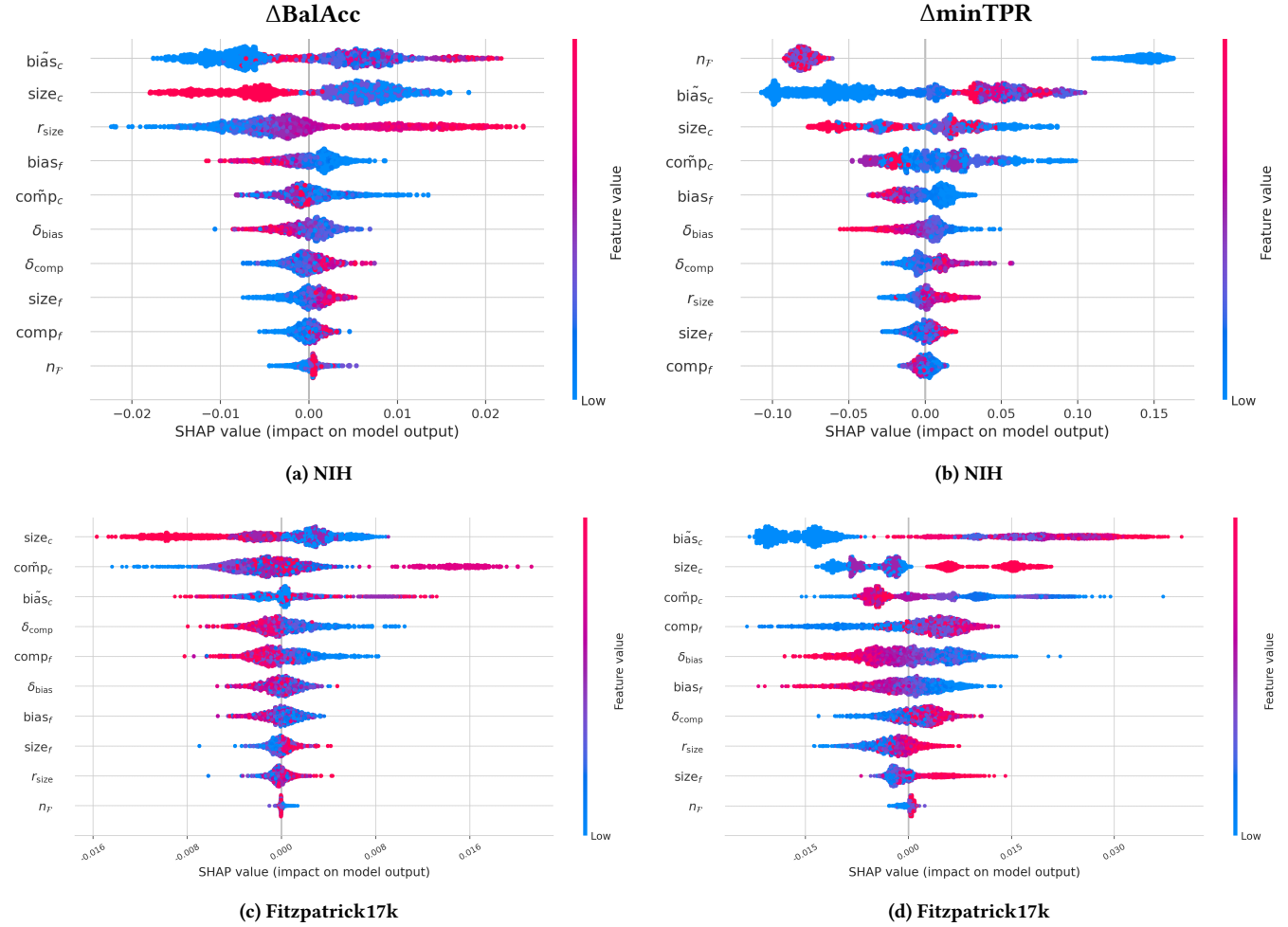


Figure 2: SHAP beeswarm plots illustrating meta-feature contributions to the predicted ΔBalAcc (left) and ΔminTPR (right) gains experienced by the candidate client upon joining a federation, for the NIH (top) and Fitzpatrick17k (bottom) datasets.

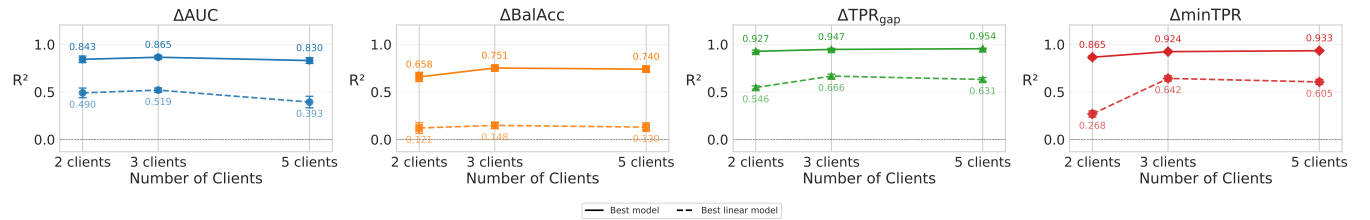


Figure 3: R^2 scores for each prediction target, evaluated separately on test points from two-, three-, and five-client federation configurations, relative to Fitzpatrick17k.

D.2 Impact on the Existing Federation Members

Predicting federation gains. Figure 4 shows actual versus predicted ΔBalAcc values and ΔminTPR values for the existing federation members upon admission of a new client, using the best-performing regression models. The R^2 values are slightly lower than those obtained from the candidate's perspective (Figure 1), yet remain satisfactory across both datasets, ranging from 0.736 to

0.860, which confirms that the federation assessment problem is well-structured and learnable also from the current members' side.

Analyzing the most predictive features. Table 2 reports the drop in R^2 for the prediction of the ΔBalAcc and ΔminTPR when features from each macro-category are removed, serving as a proxy for the relative importance of each feature family. For NIH, all three categories, size, demographic composition, and label bias,

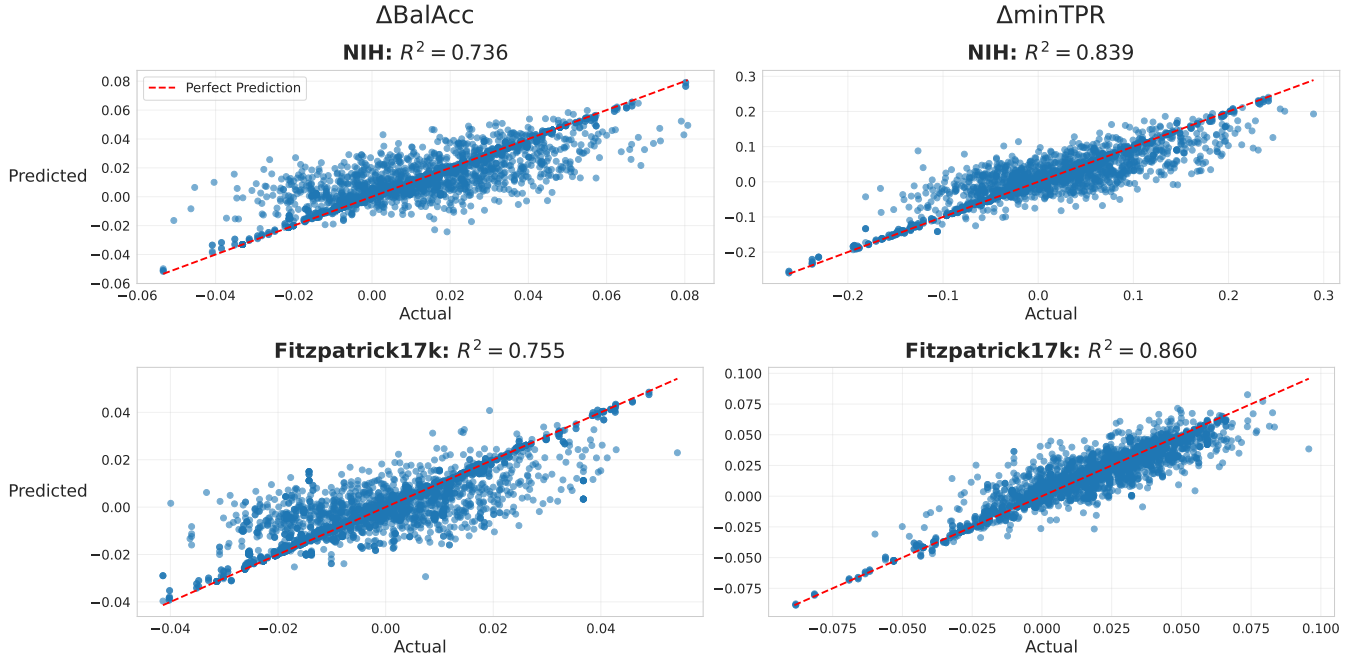


Figure 4: Actual vs. Predicted changes in BalAcc increment and minTPR increment experienced by the existing federation members upon admission of a new client, for NIH (top) and Fitzpatrick17k (bottom) datasets.

Table 2: Mean R^2 drop ($\pm$ standard deviation across five cross-validation folds) for each feature group and target, relative to NIH and Fitzpatrick17k. Results refer to the prediction of gains experienced by the current federation members upon admission of the candidate.

Feature group	NIH		Fitzpatrick17k	
	ΔBalAcc	ΔminTPR	ΔBalAcc	ΔminTPR
Size	0.178 ± 0.032	0.214 ± 0.022	0.004 ± 0.004	0.008 ± 0.006
Demographic composition	0.176 ± 0.018	0.296 ± 0.034	0.336 ± 0.020	0.201 ± 0.006
Label bias	0.270 ± 0.021	0.237 ± 0.025	0.337 ± 0.024	0.309 ± 0.021
Number of clients	0.022 ± 0.008	0.031 ± 0.010	0.000 ± 0.001	0.001 ± 0.005

contribute substantially to both targets, with label bias being the most relevant for the ΔBalAcc and demographic composition for the ΔminTPR . For Fitzpatrick17k, size-related features play a negligible role, while demographic composition and label bias contribute equally to the ΔBalAcc , and label bias alone is the strongest driver for the ΔminTPR .

Figure 5 presents the SHAP beeswarm plots for each prediction target from the federation’s perspective, relatively to NIH. For performance metrics, larger incoming clients are associated with greater gains for smaller federations, a pattern that holds for both ΔAUC and ΔBalAcc . Federations with higher internal flip fractions gain more when the candidate has lower label bias. For balanced accuracy, federations with a higher proportion of female patients show larger gains.

For fairness metrics, the directional effects differ from those observed for performance. For the TPR_{gap} reduction, federations with

more male patients benefit more, particularly when the incoming client has the opposite demographic composition. Lower label bias in the candidate is also associated with larger fairness gains for the existing federation. Relatively to the ΔminTPR , federations with higher internal label noise benefit more when the candidate has low label bias, and smaller federations show larger improvements across targets.

SHAP beeswarm plots on Fitzpatrick17k. Figure 6 presents the SHAP beeswarm plots for each prediction target on Fitzpatrick17k, from the perspective of the existing federation members. For performance metrics, gains in both AUC and balanced accuracy are more pronounced when both the candidate client and the federation contain samples from the dark-skin group, corresponding to demographic compositions of 75/25 or 50/50. Balanced accuracy

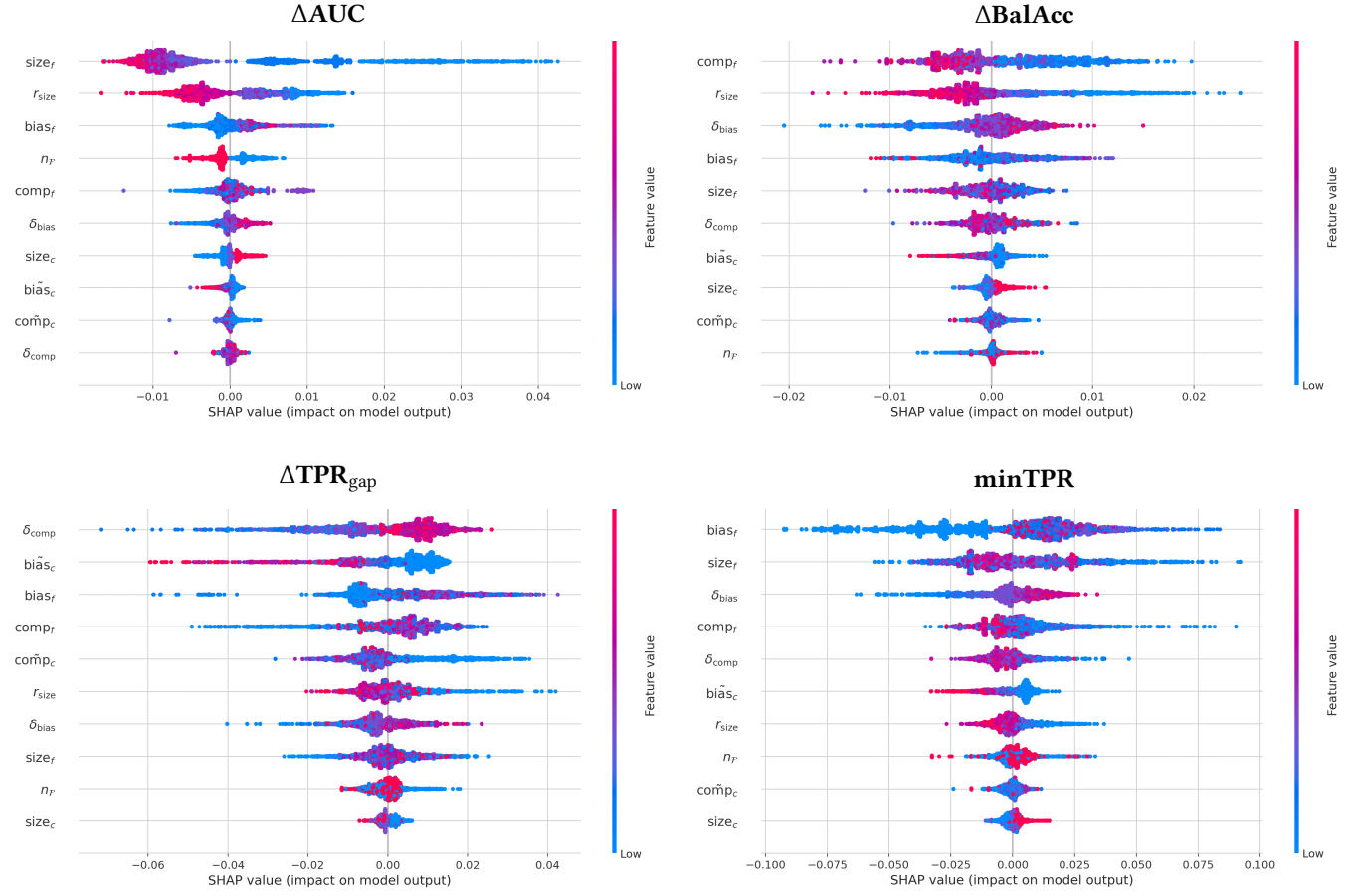


Figure 5: SHAP beeswarm plots illustrating feature contributions for each prediction target on NIH, evaluated from the perspective of the existing federation members.

gains are additionally larger when the federation is more demographically imbalanced than the candidate client, while AUC gains increase with the size of the candidate client.

For fairness metrics, improvements are stronger when the candidate client has substantially less label noise than the federation. Larger fairness gains are also observed when the federation contains considerably more light-skin samples than the candidate client. Finally, larger federations show greater fairness gains for their existing members.

Performance across federation sizes. Figure 7 shows the R^2 scores for each prediction target on Fitzpatrick17k, evaluated separately on two- and three-client configurations from the existing federation members' perspective. As observed for NIH, R^2 slightly decreases from two- to three-client configurations, reaching values of 0.822, 0.701, 0.874, and 0.842 for the three-client setting. Performance remains consistently high across both federation sizes, in line with the findings on NIH.

References

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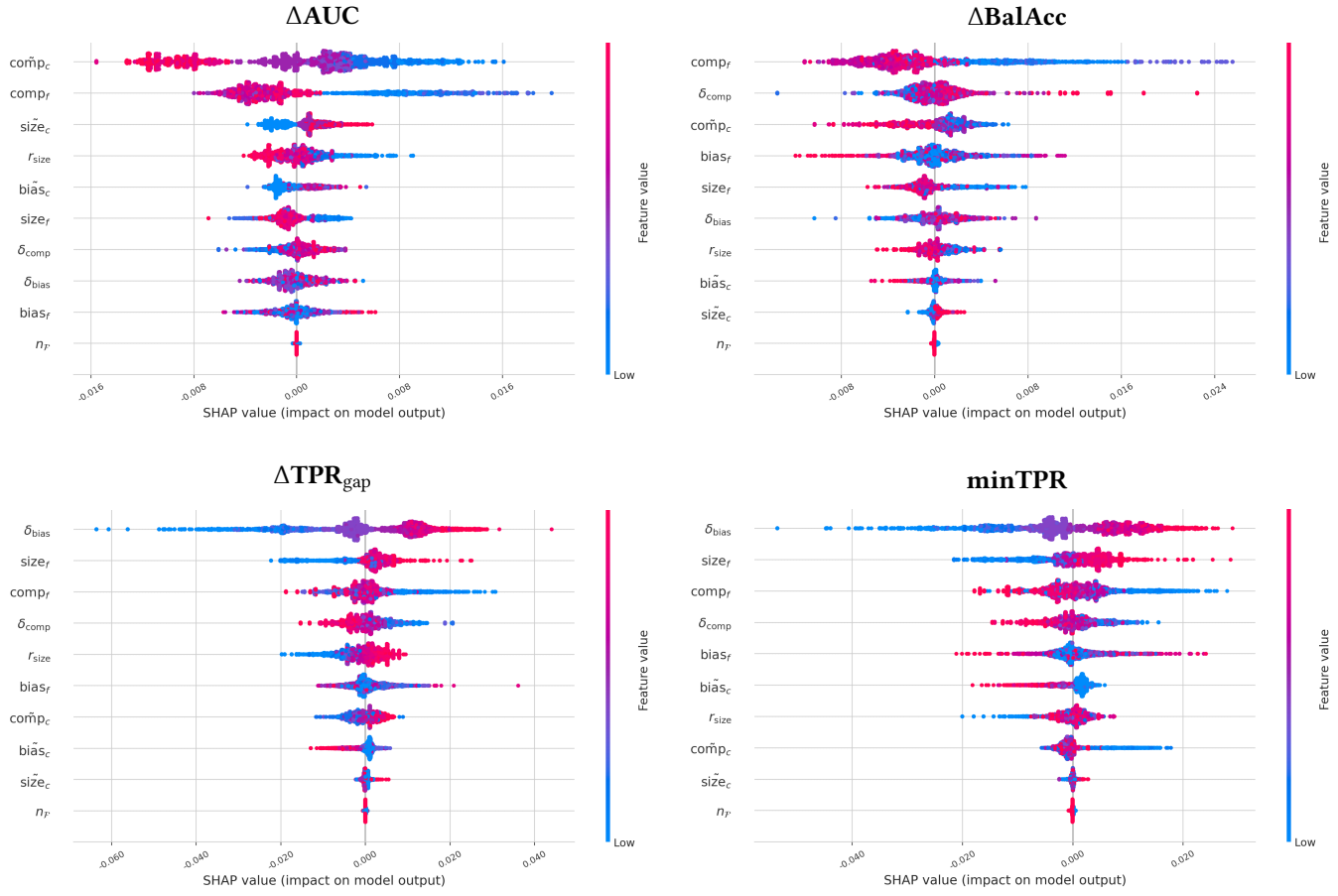


Figure 6: SHAP beeswarm plots illustrating feature contributions for each prediction target on Fitzpatrick17k, evaluated from the perspective of the existing federation members.

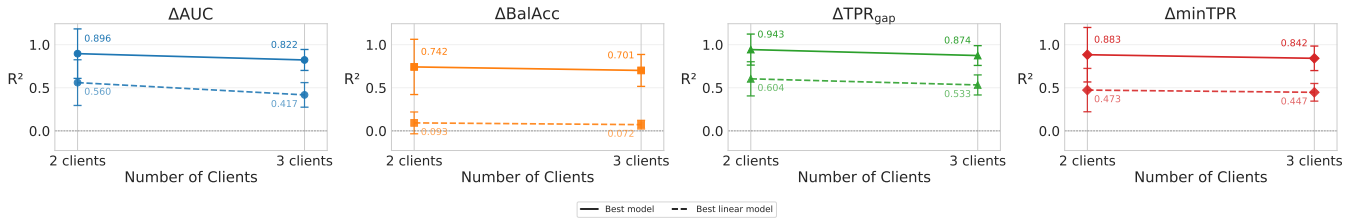


Figure 7: R^2 scores for each prediction target, evaluated separately on test points from two- and three-client federation configurations, relative to Fitzpatrick17k. Results refer to the prediction of gains experienced by the current federation members upon admission of a new client.